

What are SDN and SDWAN and What is Their Significance?

College of Science, Engineering, and Technology, Grand Canyon University

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Instructor Tim Collins

Liela Pressley

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In enterprise networks, the amount of data/network traffic that is being utilized has boomed; organizations, employees, institutions, and everyday users rely on networks to provide services, resources, data, and requests. This significant shift in relying on networks comes with the need for security to address threats and compliance, efficiency for every endpoint device and data in transit, and ease of management and configuration for network administrators. In addition, organizations are becoming heavily reliant on cloud platforms for data storage and services locally, publicly, and for WAN platforms. SDN, also known as software data networks allows network admins to control, predict, manage, and configure networks without having to manually visit each piece of hardware. WAN(wide area networks) connect users to applications and resources hosted on servers in data centers. but now as the world is becoming more cloud-centered manually configuring and managing many pieces of hardware over large distances is not ideal, efficient, and secure (*What Is SD-WAN? ,2023*). This is where SD-WAN comes in, as the SDN implementation into WAN, SDN benefits translate over to wide area networks. SDN and SDWAN provide ease of management, security, and ease of cloud integration, and are cost-effective.

SDN and SD-WAN Explained

Software-defined network, SDN, in sum, tells the hardware what to do. It is the physical separation between of network control plane and from forwarding plane. It makes decisions on how to handle traffic while data content flows between network nodes. It has central management, has a global view of the network, creates redundancy and fault tolerance, and is able to distribute policy and route changes and hardware independence. In other words, SDN provides centralized control for the admin to oversee what happens on the network and dictate

network behavior. How? It consolidates and redistributes information around the network and then makes decisions on how to change what's happening on the network for the best.

SDWAN is the application of SDN to wide-area networks. It addresses the practical realities of WAN networks and allows them to be innovative. In using SDWAN, delays are minimized across long-distance relationships of nodes, while providing reliable service with predictable quality. SDN application simplifies control and allows network decisions to be made using centralized control policies as it collects local conditions on the network. With the ease of cloud integration, traffic, and data can flow information to the cloud with full redundancy without a single point of failure. WAN can become highly scalable and upgraded without having to drastically physically change hardware (*GCU Academics, 2021*). As seen in Figure 2-0, the amount of hardware (routers) that are traditionally required within WANs is reduced because SDWAN absorbs a lot of the need for multiple devices. By software utilizing the internet and a central control over the network, administrators do not need to visit multiple hardware or tend to multiple devices to manually manage traffic.

As seen in Figure 1-0, SDWAN is the application of the same concepts utilized in SDN to benefit WAN. SDN focuses more on providing a more centralized approach for users and devices by employing APIs rather than relying on manual management. SDN provides scalability, ease of deployment, security with IoT devices, and detects/manages traffic. SD-WAN experiences the same benefits but is applied to cater to a WAN structure and its goals.

Benefits of SDN and SD-WAN

SDN and SDWAN provide ease of management, scalability, and security for network administrators across the network. By having a software-based network, updates, configurations, changes, predictions, cloud integration, etc. can be overseen with ease. It is cost-effective in a

variety of ways, software allows interoperability across a variety of generic hardware across the network. It is more expensive to rely on the traditional methods or inefficiently use backup circuits and not address the multi-connectivity between users and devices and cloud environments. SD-WAN allows companies to blend WAN into many different cloud platforms such as public clouds, Microsoft Office 365, Salesforce, AWS, Microsoft Azure, etc. In addition, it reduces costs by enabling transport independence across MPLS (Multiprotocol Label Switching), 4G/5G LTE, and other connections. Lastly, it reduces costs by enabling optimized user experience and efficiency for software-as-a-service and public cloud applications(*What Is SD-WAN?*,2023).

The heavy amount of traffic that WANs experience needs to be managed proficiently in order to confront the complexity, and predict application performance and the vulnerabilities in data (in transit or at rest). With the boom of using the Internet and cloud for data storage and access, there are many threats and compliance issues that need to be addressed. SD-WAN offers secure traffic across the broadband Internet and to the cloud, distributes security protocols to the remote endpoints, and has application-aware policies that can be implemented end-to-end with real access control(*What Is SD-WAN?*,2023).

Prerequisites/Planning Software-Defined Network Infrastructure

There is a precise list of requirements network administrators need to review prior to implementing any form of SDN infrastructure. This list of prerequisites includes security groups and dynamic DNS registration, having access to physical network devices in order to configure VLANs, routing, and BGP, and having physical compute hosts to run Hyper-V. Each physical compute host needs network connectivity via 1 or more network adapters to a physical switch port. Ensure that switches and routers match specific hardware requirements such as having L3

protocols, BGP (border gateway protocol), equal-cost multi-path routing, and more
(AnirbanPaul,2023).

Illustrations

Figure 1-0 SDN and SDWAN (BNumeryx, n.d.)

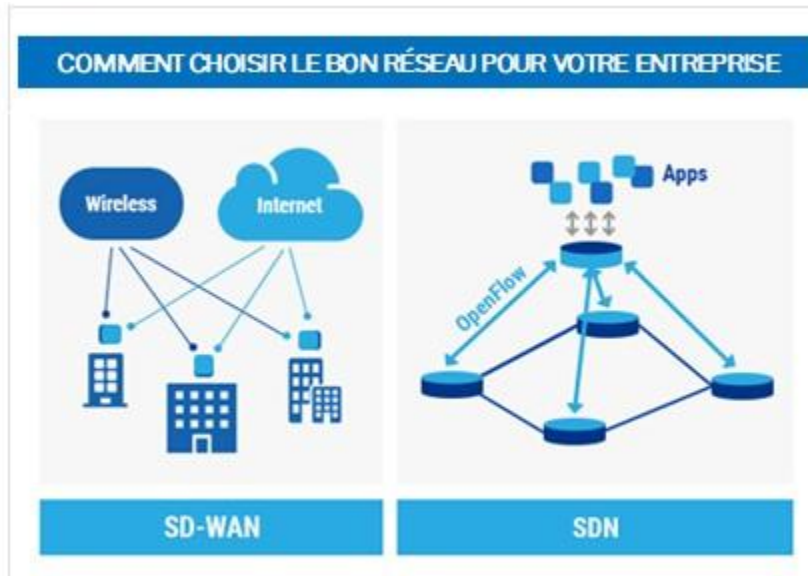
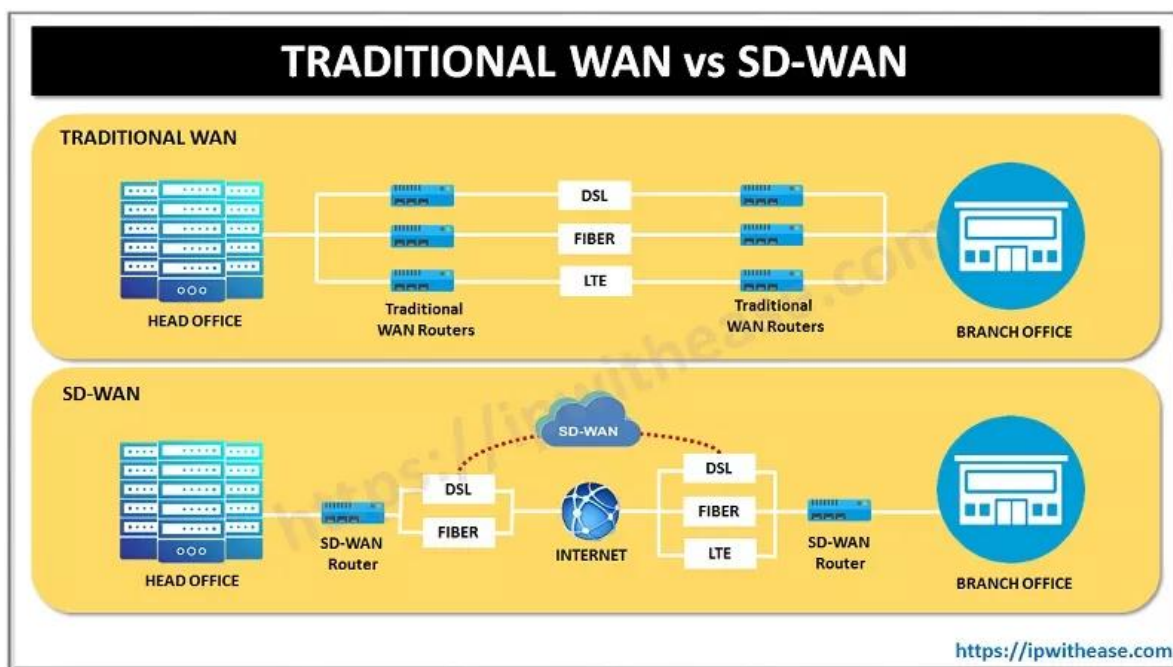


Figure 2-0- WAN vs SD-WAN (Bhardwaj,2021)



References

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